GSE 518
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Problem Set 10

You do not need to hand in this problem set, but the material covered in it will appear in the final exam. Please let me know if you have any questions

1. Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution $N(\mu_0, \sigma_0^2)$. Let $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$. Show s^2 is consistent.

2. Problem 10.1 in CB. (*Hint:* Try the method of moments estimator. Also, just to save you some time, you can take as given that $E(X) = \frac{\theta}{3}$ and $Var(X) = \frac{1}{3} - \frac{\theta_0^2}{9}$.)

3. Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution $N(\mu, \sigma_0^2)$, where μ is *known* and σ_0^2 is *unknown*. In class we obtained that the MLE for σ_0^2 is $\widehat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$ and its limiting distribution is given by

$$\sqrt{n}(\widehat{\sigma}^2_{MLE} - \sigma_0^2) \xrightarrow{d} N\left(0, 2\sigma_0^4\right)$$

   (a) Find the maximum likelihood estimator for the standard deviation, σ_0.

   (b) Find the asymptotic distribution of your MLE estimator for σ_0. (*Hint:* Think about the Delta Method.)

4. Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with pdf

$$f(x|\theta_0) = \theta_0 x^{\theta_0 - 1}$$

   for $0 < x < 1$, and zero otherwise, for $0 < \theta_0 < \infty$.

   (a) Find an estimator for θ_0 which you know is consistent, asymptotically normally distributed and asymptotically efficient. (*Hint:* Check a very nice theorem seen in class.)

   (b) Obtain the asymptotic distribution of your estimator in part (a).

   (c) Based on your result in part (b), how would you estimate the variance of your estimator in part (a)? (*Hint:* Think about its MLE estimator.)

5. Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with pdf $f(x|\theta_0) = \theta_0 e^{-\theta_0 x}$, for $0 < x < \infty$, and zero otherwise, for $0 < \theta_0 < \infty$. Find the maximum likelihood estimator for θ_0 and derive its asymptotic distribution.

6. Let $X_1, X_2, \ldots, X_n$ be a random sample from a Poisson distribution with pmf $f(x|\theta_0) = \frac{\theta_0^x e^{-\theta_0}}{x!}$, for $x = 0, 1, \ldots$ and zero otherwise, with $\theta_0 \geq 0$.

   (a) Find the maximum likelihood estimator for θ_0, $\widehat{\theta}_{MLE}$, and derive the limiting distribution of $\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0)$.

   (b) Suppose we want to estimate the probability that $X = 0$: $\gamma_0 = \Pr(X = 0) = e^{-\theta_0}$. Find the maximum likelihood estimator for γ_0 and derive its asymptotic distribution.